PyCharm File Edit View Navigate Code Refactor Run Tools VCS Window Help Tue 28 Jul 1:43 PM

PyCharmMiscProject Version control main Start Free Trial

Exp 13.py

```python
grammar = {'S': [['NP', 'VP']],'NP': [['Det', 'N']],'VP': [['V', 'NP']],'Det': [['the'], ['a']],'N': [['cat'], ['dog']],'V': [['sees'], ['likes']]}
sentence = input("Enter sentence: ").lower().split()
def parse(symbol, words, pos):  2 usages
    if symbol not in grammar:
        if pos < len(words) and symbol == words[pos]:
            return (symbol, []), pos + 1
        return None, pos
    for rule in grammar[symbol]:
        children = []
        current = pos
        success = True
        for item in rule:
            tree, current = parse(item, words, current)
            if tree is None:
                success = False
                break
            children.append(tree)
        if success:
            return (symbol, children), current
    return None, pos
def print_tree(tree, level=0):  2 usages
    symbol, children = tree
    print("  " * level + symbol)
    for child in children:
        print_tree(child, level + 1)
tree, position = parse( symbol: 'S', sentence,  pos: 0)
```

Run main

```
Process finished with exit code 1
```

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Exp 13.py

```python
    if tree and position == len(sentence):
        print("\nParse Tree:")
        print_tree(tree)
    else:
        print("Sentence cannot be parsed")
```

Run: main

```
/Users/moshika/PyCharmMiscProject/.venv/bin/python /Users/moshika/PyCharmMiscProject/Exp 13.py
Enter sentence: the cat sees a dog

Parse Tree:
S
  NP
    Det
      the
    N
      cat
  VP
    V
      sees
    NP
      Det
        a
      N
        dog

Process finished with exit code 0
```